

Model Predictive Control for Active Rear Steering Without Sideslip-Angle Feedback

Myeongseok Ryu, Soobin Hwang, and Kyunghwan Choi

Abstract—This paper proposes a model predictive control (MPC) method for active rear steering (ARS) that does not require sideslip-angle feedback. ARS control can improve vehicle maneuverability and lateral stability while preserving the driver’s steering intention through the front wheels. However, conventional ARS controllers commonly rely on the vehicle sideslip angle, which must be obtained through direct sensing or online estimation and may therefore introduce sensing or estimation errors into the feedback loop. To eliminate this dependence, the proposed MPC employs two complementary mechanisms. First, the yaw rate is predicted without using the sideslip angle by exploiting its limited influence on finite-horizon yaw-rate prediction under the considered operating conditions. Second, sideslip-angle attenuation is promoted by suppressing a residual term in the sideslip-angle dynamics that is independent of the sideslip angle itself. These two mechanisms enable the MPC optimization to be formulated using yaw-rate and steering signals without requiring measured or estimated sideslip-angle feedback. Numerical simulations demonstrate that the proposed controller maintains satisfactory yaw-rate tracking and sideslip suppression while remaining insensitive to sideslip-angle feedback errors.

I. INTRODUCTION

Steering control plays a crucial role in vehicle dynamics control, as it directly generates lateral forces that affect the vehicle’s maneuverability and stability [1]. In this manner, four wheel steering (4WS) vehicles maximize the steering capability by utilizing both front and rear wheels, compared to conventional front wheel steering (FWS) vehicles. Recently, active four wheel steering (A4WS) control, which actively controls all four wheels, has been widely studied for autonomous driving [2]. However, for production vehicles that are primarily human-driven, active rear steering (ARS) control is more compatible than A4WS control because the front-wheel steering remains directly governed by the driver, thereby preserving the driver’s steering intention [3].

Many control methods have been adopted for ARS control. The following review covers both stand-alone ARS control and ARS control combined with other controls. In early studies conducted until the 1990s, proportional control to the longitudinal velocity [4] and front steering angle [5] were proposed. As the electronic control technology has been developed, more advanced control methods have been proposed, such as sliding mode control (SMC) [6], [7] and

H_∞ control [8]. These control methods have successfully shown the expected control behavior of ARS control [9], which is to steer the rear wheels in the opposite (same) direction of the front wheels at low (high) speeds to improve maneuverability (stability).

The control performance of these methods is generally satisfactory; however, the phase lag between sideslip angle and yaw rate remains a limitation [10]. Since the sideslip angle is initially small when the vehicle starts to steer, the controller tends to prioritize yaw-rate tracking, resulting in an excessively aggressive initial rear-wheel steer input. In this context, infinite-horizon linear quadratic regulator (LQR) [11] and finite-horizon model predictive control (MPC) [3], [12] have been proposed. These approaches provide a systematic framework for balancing the respective control objectives for yaw rate and sideslip angle by accounting for the vehicle-state evolution over their design horizons.

Meanwhile, because the vehicle sideslip angle is difficult to measure directly, practical implementations commonly rely on estimation methods such as extended Kalman filters (EKF) [13], sliding-mode observers [14], and Luenberger observers [15]. However, both direct sensing and model-based estimation are subject to sensing noise, modeling uncertainty, and estimation errors. When the estimated sideslip angle is directly fed back to the controller, these errors can propagate into the control input and degrade the closed-loop performance. This motivates an ARS control design that does not require sideslip-angle feedback itself, rather than relying on increasingly accurate sideslip-angle estimation.

In this paper, we propose an MPC-based ARS controller that removes reliance on sideslip-angle feedback. The proposed design removes the dependence on the sideslip angle at the two locations where it is conventionally used in the MPC formulation. First, a sideslip-angle-independent yaw-rate predictor is constructed by exploiting the limited finite-horizon sensitivity of the predicted yaw rate to the initial sideslip angle. Second, instead of directly penalizing the sideslip angle, the controller suppresses the residual term in the sideslip-angle dynamics, thereby promoting sideslip attenuation without using sideslip-angle feedback. These two mechanisms eliminate the need for either direct sideslip-angle measurement or online sideslip-angle estimation in the feedback loop. The main contribution is therefore a structurally sideslip-angle-feedback-free MPC formulation that retains yaw-rate tracking and indirect sideslip stabilization.

Notation: $\mathbb{R}$ and $\mathbb{N}$ denote the set of real numbers and natural numbers, respectively. For $\mathbf{x}_i \in \mathbb{R}^{m_i \times p}$, $\forall i \in \{1, \dots, n\}$, let $[\mathbf{x}_i]_{i=1}^n$ denote their vertical concatenation, i.e., $[\mathbf{x}_i]_{i=1}^n :=$

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TABLE I: Vehicle Parameters.

Symbol	Description	Value
m	Mass	2335.07 kg
I_z	Yaw moment of inertia	5376.432 kg m ²
l_f	Distance from C.G. to front axle	1.574 m
l_r	Distance from C.G. to rear axle	1.566 m
C_f	Cornering stiffness of front tires	119.54 kN rad ⁻¹
C_r	Cornering stiffness of rear tires	119.83 kN rad ⁻¹

$[\mathbf{x}_1^\top, \dots, \mathbf{x}_n^\top]^\top \in \mathbb{R}(\sum_{i=1}^n m_i) \times p$. The entry of $\mathbf{A} \in \mathbb{R}^{m \times n}$ in the i -th row and j -th column is denoted by $[\mathbf{A}]_{(i,j)}$. The Gaussian white noise with zero mean and variance σ^2 is denoted by $\mathcal{N}(0, \sigma^2)$. The Kronecker product is denoted by $\otimes$. The $n \times n$ identity matrix and the n -dimensional column vector of ones are denoted by $\mathbf{I}_n$ and $\mathbf{1}_n$, respectively.

The remainder of this paper is organized as follows. Section II describes the system modeling and control objective. Section III presents the control-oriented analyses of the lateral vehicle dynamics. Section IV presents the proposed MPC design by introducing alternative objective functions that do not require sideslip-angle feedback. Section V validates the proposed ARS control design in numerical simulations. Finally, Section VI concludes the paper.

II. SYSTEM MODELING AND CONTROL OBJECTIVE

We consider a simple 2-degree-of-freedom (2-DOF) bicycle model which is presented in [1, Sec. 2.6], as follows:

$$\frac{d}{dt} \begin{pmatrix} \beta \\ r \end{pmatrix} = \begin{pmatrix} \frac{1}{mv_x} (F_{y,f} \cos(\delta_f) + F_{y,r} \cos(\delta_r)) - r \\ \frac{1}{I_z} (l_f F_{y,f} \cos(\delta_f) - l_r F_{y,r} \cos(\delta_r)) \end{pmatrix}, \quad (1)$$

where β, r and v_x denote the sideslip angle, yaw rate, and longitudinal velocity of the vehicle, respectively, and the other parameters are defined in Table I. The front and rear steering angles are denoted by δ_f and δ_r , respectively. The lateral tire force $F_{y,i}$ is a nonlinear function of the tire slip angle α_i , vertical loads, temperature, and other factors for the front ($i = f$) and rear ($i = r$) tires. We employ a linear tire model which is given by

$$F_{y,i} = C_i \alpha_i, \quad \forall i \in \{f, r\}. \quad (2)$$

Using the small-angle approximation, the slip angles can be approximated as

$$\alpha_f \approx \delta_f - \left(\beta + \frac{l_f r}{v_x} \right), \quad \alpha_r \approx \delta_r - \left(\beta - \frac{l_r r}{v_x} \right), \quad (3)$$

which are typically valid for $|\alpha_i| \leq 5^\circ$, $\forall i \in \{f, r\}$.

Accordingly, the vehicle dynamics (1) can be rewritten as a linear system:

$$\frac{d}{dt} \mathbf{x} = \mathbf{A} \mathbf{x} + \mathbf{b}_f \delta_f + \mathbf{b}_r \delta_r, \quad (4)$$

where $\mathbf{x} := (\beta, r)^\top$ denotes the state, and the system matrices $\mathbf{A}$ and control effectiveness vectors, $\mathbf{b}_f, \mathbf{b}_r$, are

defined as

$$\mathbf{A} = \begin{bmatrix} -\frac{C_f + C_r}{mv_x} & \frac{C_r l_r - C_f l_f}{mv_x^2} - 1 \\ -\frac{C_f l_f - C_r l_r}{I_z} & -\frac{C_f l_f^2 + C_r l_r^2}{I_z v_x} \end{bmatrix}, \quad (5)$$

and

$$\mathbf{b}_f = \begin{pmatrix} \frac{C_f}{mv_x} \\ \frac{C_f l_f}{I_z} \end{pmatrix}, \quad \mathbf{b}_r = \begin{pmatrix} \frac{C_r}{mv_x} \\ -\frac{C_r l_r}{I_z} \end{pmatrix}. \quad (6)$$

The reference yaw rate trajectory is generated using the following equation adopted from [3]:

$$r_d(\delta_f) = \min \left\{ \left| \frac{v_x \delta_f}{(L + K v_x^2)} \right|, \left| \frac{0.85 \mu g}{v_x} \right| \right\} \text{sgn}(\delta_f), \quad (7)$$

where $L := l_f + l_r$ is the wheelbase, μ is the road friction coefficient, $g = 9.81$ is the gravitational acceleration, and K is the understeer gradient defined as $K = \frac{m}{L} \left(\frac{l_r}{C_f} - \frac{l_f}{C_r} \right)$. It should be noted that the reference sideslip angle is not considered because the sideslip angle is not measurable; therefore, exact reference tracking is inherently difficult.

For the predictive control, the system (4) is discretized with sampling time $T_s \in \mathbb{R}_{>0}$ using Euler discretization, as follows:

$$\mathbf{x}_{k+1} = \mathbf{A}_d \mathbf{x}_k + \mathbf{b}_{f,d} \delta_{f,k} + \mathbf{b}_{r,d} \delta_{r,k}, \quad (8)$$

where $\mathbf{A}_d := \mathbf{I}_2 + \mathbf{A} T_s$ and $\mathbf{b}_{i,d} := \mathbf{b}_i T_s$, $\forall i \in \{f, r\}$, and $k \in \mathbb{N}$ is the discrete-time index. Each state and control input at the k -th time step is denoted using the subscript k .

The control objective is to generate the ARS control input δ_r for a given driver's front-steering input δ_f to improve vehicle maneuverability and stability. In particular, we assume that the sideslip angle is not measurable; therefore, the controller should be designed without sideslip-angle feedback.

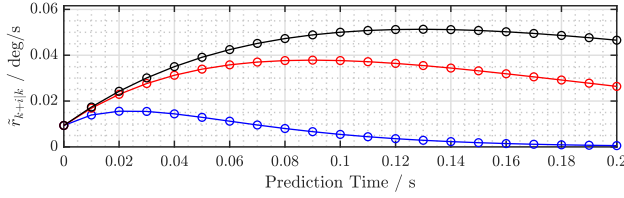
III. CONTROL-ORIENTED ANALYSIS FOR MPC WITHOUT SIDESLIP-ANGLE FEEDBACK

The proposed MPC eliminates the need for sideslip-angle feedback through two complementary mechanisms: sideslip-angle-independent yaw-rate prediction and indirect sideslip stabilization through residual suppression. This section provides the control-oriented analyses underlying these two mechanisms. Unless otherwise stated, all analyses are conducted using the vehicle parameters listed in Table I under normal driving conditions over the longitudinal-speed range $v_x \in (0, 110]$, km h⁻¹.

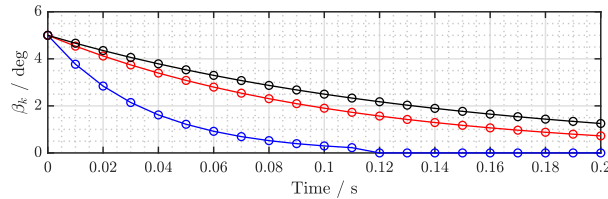
A. Limited Influence of Sideslip Angle on Finite-Horizon Yaw-Rate Prediction

The conventional MPC prediction requires the full initial state $\mathbf{x}_k = (\beta_k \ r_k)^\top$. Because β_k is unavailable without measurement or estimation, we first examine how strongly the finite-horizon yaw-rate prediction depends on the initial sideslip angle. The following analysis shows that this dependence is sufficiently small under the considered operating conditions, motivating a yaw-rate predictor that does not use β_k .

$$\Phi := \begin{bmatrix} \mathbf{A}_d \\ \mathbf{A}_d^2 \\ \vdots \\ \mathbf{A}_d^{N_p} \end{bmatrix}, \quad \Gamma_f := \begin{pmatrix} \mathbf{b}_{f,d} \\ \mathbf{A}_d \mathbf{b}_{f,d} \\ \vdots \\ \sum_{i=1}^{N_p} \mathbf{A}_d^{i-1} \mathbf{b}_{f,d} \end{pmatrix}, \quad \Gamma_r := \begin{bmatrix} \mathbf{b}_{r,d} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{A}_d \mathbf{b}_{r,d} & \mathbf{b}_{r,d} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_d^{N_p-1} \mathbf{b}_{r,d} & \mathbf{A}_d^{N_p-2} \mathbf{b}_{r,d} & \cdots & \sum_{i=0}^{N_p-N_c} \mathbf{A}_d^i \mathbf{b}_{r,d} \end{bmatrix} \quad (10)$$



(a) Yaw-rate prediction error with an initial sideslip-angle error of 5 deg.

(b) Sideslip-angle response under residual suppression with the initial condition $\beta_k = 5$ deg.Fig. 1: Control-oriented analysis results at $v_x = 30 \text{ km h}^{-1}$ [—○—], 80 km h^{-1} [—●—], and 110 km h^{-1} [—●—].

The state sequence over the prediction horizon is defined as

$$\mathbf{X}_k = \Phi \mathbf{x}_k + \Gamma_f \delta_{f,k} + \Gamma_r \mathbf{U}_k, \quad (9)$$

where $\Phi \in \mathbb{R}^{2N_p \times 2}$, $\Gamma_f \in \mathbb{R}^{2N_p \times 1}$, and $\Gamma_r \in \mathbb{R}^{2N_p \times N_c}$ are the prediction matrices defined in (10), with $N_p, N_c \in \mathbb{N}$ being the prediction and control horizons, respectively, where $N_c \leq N_p$. The state sequence predicted over N_p from the initial state $\mathbf{x}_k$ is defined as $\mathbf{X}_k = [\mathbf{x}_{k+i|k}]_{i=1}^{N_p} \in \mathbb{R}^{2N_p}$, where $\mathbf{x}_{k+i|k} \in \mathbb{R}^2$ denotes the state at time step $k+i$ predicted using $\mathbf{x}_k$, for all $i \in \{1, \dots, N_p\}$. The front-wheel steer angle is assumed to remain constant over the prediction horizon. The rear-wheel steer-angle sequence over the control horizon N_c is defined as $\mathbf{U}_k := [u_{k,i}]_{i=0}^{N_c-1} \in \mathbb{R}^{N_c}$. Beyond the control horizon, the rear-wheel steer angle is held at the final value of the control sequence over the remainder of the prediction horizon, i.e., $u_{k,i} = u_{k,N_c-1}$, $\forall i \in \{N_c, \dots, N_p-1\}$. The sequence $\mathbf{U}_k$ serves as the optimization variable in the MPC formulation presented in Section IV.

Assume that the initial state used for prediction is constructed using an estimated sideslip angle $\hat{\beta}_k$, i.e., $\hat{\mathbf{x}}_k := (\hat{\beta}_k \ r_k)^\top$. The resulting prediction error can then be quantified as follows:

$$\mathbf{X}_k - \hat{\mathbf{X}}_k = \Phi(\mathbf{x}_k - \hat{\mathbf{x}}_k) = \Phi \begin{pmatrix} \beta_k - \hat{\beta}_k \\ 0 \end{pmatrix}, \quad (11)$$

where $\hat{\mathbf{X}}_k$ denotes the predicted state sequence from $\hat{\mathbf{x}}_k$ using (9). Then, the yaw-rate prediction error at time step $k+i$, denoted by $\tilde{r}_{k+i|k}$, is obtained by extracting the

corresponding yaw-rate component from $\mathbf{X}_k - \hat{\mathbf{X}}_k$ in (11), as follows:

$$\tilde{r}_{k+i|k} := r_{k+i|k} - \hat{r}_{k+i|k} = [\mathbf{A}_d^i]_{(2,1)}(\beta_k - \hat{\beta}_k), \quad (12)$$

where $r_{k+i|k}$ and $\hat{r}_{k+i|k}$ denote the predicted yaw rate with $\mathbf{x}_k$ and $\hat{\mathbf{x}}_k$ at time step $k+i$, respectively.

In normal driving conditions, the sideslip angle is typically limited within 5 deg, which is used as the initial condition error in this analysis. The prediction error of yaw rate $\tilde{r}_{k+i|k}$ is analyzed with $N_p = 20$ and $T_s = 0.01$ s. As shown in Fig. 1a, the prediction error of yaw rate increases with respect to the longitudinal velocity, but ultimately decreases along the prediction horizon. Notably, even with an initial sideslip-angle error of 5 deg, the yaw-rate prediction error remains below 0.05 deg/s at $v_x = 110 \text{ km h}^{-1}$, corresponding to less than 0.5% of the typical yaw-rate magnitude under normal driving conditions (10 deg/s to 20 deg/s).

This result indicates that the yaw-rate prediction is not significantly affected by the initial sideslip-angle error, and thus, the sideslip angle can be neglected in the yaw-rate prediction, as follows:

$$r_{k+1|k} \approx [\mathbf{A}_d]_{(2,2)} r_k + [\mathbf{b}_{f,d}]_{(2,1)} \delta_{f,k} + [\mathbf{b}_{r,d}]_{(2,1)} \delta_{r,k}. \quad (13)$$

Extending the one-step approximation in (13) over the prediction horizon, the predicted yaw-rate sequence can be approximately expressed by omitting β_k from the initial state in (9), as follows:

$$\mathbf{r}_{\text{pred}|k} \approx \phi^{(2,2)} r_k + \gamma_f^{(r)} \delta_{f,k} + \gamma_r^{(r)} \mathbf{U}_k, \quad (14)$$

where $\phi^{(2,2)} := \mathbf{S}_r \Phi (0 \ 1)^\top$, $\gamma_i^{(r)} := \mathbf{S}_r \Gamma_i$, $\forall i \in \{f, r\}$, with $\mathbf{S}_r := (\mathbf{I}_{N_p} \otimes (0 \ 1))$.

B. Residual Suppression for Sideslip Angle Stabilization

Since the sideslip angle is difficult to measure in real time, we indirectly stabilize it by suppressing the residual term, motivated by [10]. First, the sideslip-angle dynamics in (8) can be rewritten as

$$\beta_{k+1} = [\mathbf{A}_d]_{(1,1)} \beta_k + \underbrace{[\mathbf{A}_d]_{(1,2)} r_k + [\mathbf{b}_{f,d}]_{(1)} \delta_{f,k} + [\mathbf{b}_{r,d}]_{(1)} \delta_{r,k}}_{=: \Delta_k(r_k, \delta_{f,k}, \delta_{r,k})}, \quad (15)$$

where $\Delta_k := \Delta_k(r_k, \delta_{f,k}, \delta_{r,k})$ denotes the residual term. By canceling Δ_k with the rear-wheel steer input $\delta_{r,k}$, the sideslip-angle dynamics can be stabilized with the stable pole $[\mathbf{A}_d]_{(1,1)}$, whose magnitude is less than one under the considered conditions.

Extending the one-step residual-suppression argument over the prediction horizon, the predicted sideslip sequence

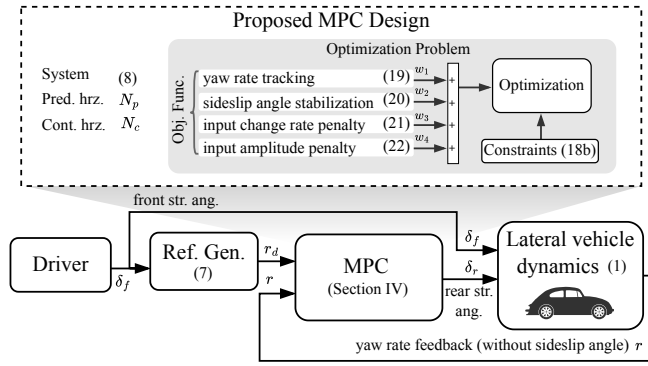


Fig. 2: Proposed MPC design for ARS control.

can be expressed as

$$\beta_{\text{pred}|k} = \phi^{(1,1)}\beta_k + \underbrace{\phi^{(1,2)}r_k + \gamma_f^{(\beta)}\delta_{f,k} + \gamma_r^{(\beta)}U_k}_{:=\Delta_{\text{pred}|k}(r_k, \delta_{f,k}, U_k)}, \quad (16)$$

where $\Delta_{\text{pred}|k} := \Delta_{\text{pred}|k}(r_k, \delta_{f,k}, U_k)$ denotes the residual term along the prediction horizon, and $\phi^{(1,1)} := S_\beta \Phi \begin{pmatrix} 1 & 0 \end{pmatrix}^\top$, $\phi^{(1,2)} := S_\beta \Phi \begin{pmatrix} 0 & 1 \end{pmatrix}^\top$, $\gamma_i^{(\beta)} := S_\beta \Gamma_i$, $\forall i \in \{f, r\}$, with $S_\beta := (I_{N_p} \otimes \begin{pmatrix} 1 & 0 \end{pmatrix})$.

All magnitudes of the entries of $\phi^{(1,1)}$ are less than one under the considered conditions, which indicates that the future sideslip angles are smaller in magnitude than the initial sideslip angle, if $\Delta_{\text{pred}|k}$ is sufficiently suppressed. This attenuation is illustrated in Fig. 1b for $\beta_k = 5^\circ$ under ideal residual suppression, i.e., $\Delta_{\text{pred}|k} = 0$. However, the entries of $\phi^{(1,1)}$ are inversely proportional to the longitudinal velocity, thereby decreasing the convergence rate as the longitudinal velocity increases. Quantitatively, the settling time from a given initial sideslip angle to 5% of its initial value can be obtained by

$$t_{5\%} = T_s \cdot \min_i \left\{ i \in \mathbb{N} : \left| [A_d^i]_{(1,1)} \right| \leq 0.05 \right\}. \quad (17)$$

Under the considered conditions, the settling time is 0.11 s and 0.44 s at $v_x = 30 \text{ km h}^{-1}$ and $v_x = 110 \text{ km h}^{-1}$, respectively. This demonstrates that residual suppression can effectively stabilize the sideslip-angle dynamics even at high longitudinal velocities. Nevertheless, it should be noted that the residual term has to be properly suppressed to achieve the anticipated performance, especially at higher speeds.

IV. PROPOSED MPC DESIGN FOR ARS CONTROL

In this section, we present the proposed MPC design for ARS control, as illustrated in Fig. 2. Based on the control-oriented analyses in Section III, the proposed MPC optimization problem is formulated as follows:

$$\min_{U_k} J(U_k) := \sum_{i=1}^4 w_i J_i(U_k) \quad (18a)$$

$$\text{subject to} \quad -1_{N_c} \bar{\delta}_r \leq U_k \leq 1_{N_c} \bar{\delta}_r, \quad (18b)$$

where $w_i \in \mathbb{R}_{>0}$ is the weight assigned to objective function $J_i := J_i(U_k)$ for each $i \in \{1, 2, 3, 4\}$, and $\bar{\delta}_r$ is the maximum rear steering angle. The first two terms J_1

and J_2 address maneuverability and stability, respectively. Here, J_1 employs the sideslip-angle-independent yaw-rate prediction in (14), whereas J_2 penalizes the residual of the sideslip-angle dynamics in (16), allowing both objectives to be evaluated without requiring sideslip-angle feedback. The remaining terms J_3 and J_4 are designed for control effort and smoothness, respectively.

1) *Objective Functions for Maneuverability and Stability:* For maneuverability, J_1 is designed to track the reference trajectory of yaw rate, as follows:

$$J_1(U_k) := \frac{1}{2} \|\mathbf{r}_{\text{pred}|k} - \mathbf{r}_{d,k}\|^2, \quad (19)$$

where $\mathbf{r}_{d,k} = 1_{N_p} \cdot r_d(\delta_{f,k})$ is a constant reference yaw-rate trajectory over the prediction horizon (see, (7)), and $\mathbf{r}_{\text{pred}|k}$ is the approximately predicted sequence of yaw rate defined in (14) without sideslip-angle feedback. On the other hand, for stability, J_2 is designed to suppress the residual term $\Delta_{\text{pred}|k}$ in (16), as follows:

$$J_2(U_k) := \frac{1}{2} \|\Delta_{\text{pred}|k}\|^2. \quad (20)$$

Together, the two mechanisms remove the sideslip angle from both the yaw-rate prediction and the stability-related objective. Consequently, the MPC optimization can be evaluated using the yaw rate and front steering signals only, without requiring either measured or estimated sideslip-angle feedback.

On the other hand, the weights w_1 and w_2 have a trivial trade-off relationship, which can be explained as follows. If the importance of J_1 is emphasized by increasing w_1 , the rear wheel steering input δ_r is generated to track the reference trajectory of yaw rate, which may allow the residual term to remain large, thereby destabilizing the sideslip-angle dynamics. On the other hand, if the importance of J_2 is emphasized by increasing w_2 , the tracking performance of yaw rate may be degraded. Thus, w_1 and w_2 should be carefully tuned to achieve a desired trade-off between maneuverability and stability, especially according to operating conditions, such as longitudinal velocity and road friction coefficient.

2) *Objective Functions for Control Input:* The remaining objective functions J_3 and J_4 are designed to penalize the control effort and smoothness, respectively, which are commonly used in MPC design. First, J_3 is designed as follows:

$$J_3(U_k) := \frac{1}{2} \|U_k\|^2. \quad (21)$$

On the other hand, J_4 is designed to penalize the difference between consecutive control inputs, as follows:

$$J_4(U_k) := \frac{1}{2} \|\mathbf{D}U_k - \mathbf{u}_{\text{prev}}\|^2, \quad (22)$$

where $\mathbf{D} \in \mathbb{R}^{N_c \times N_c}$ is defined as

$$[\mathbf{D}]_{(i,j)} = \begin{cases} 1, & i = j \\ -1, & i = j + 1 \\ 0, & \text{otherwise} \end{cases}, \quad (23)$$

and $\mathbf{u}_{\text{prev}} := (\delta_{r,k-1} \ 0 \ \cdots \ 0)^\top \in \mathbb{R}^{N_c}$.

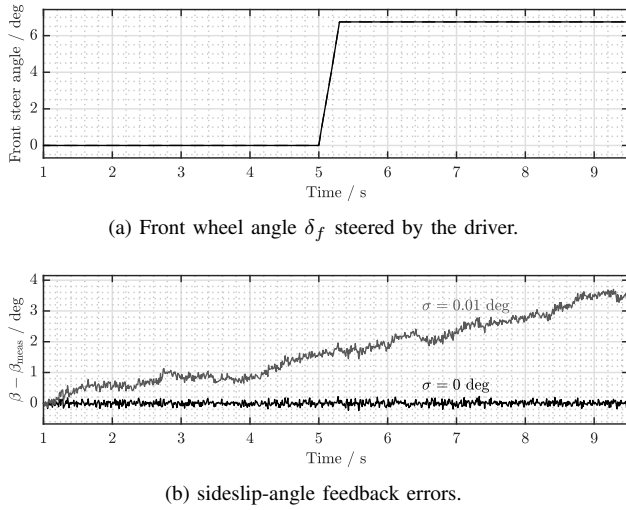


Fig. 3: Scenario setup for the validations.

The MPC problem in (18) can be reformulated as a standard quadratic programming (QP) problem, as follows:

$$\min_{\mathbf{U}_k} \quad \frac{1}{2} \mathbf{U}_k^\top \mathbf{H} \mathbf{U}_k + \mathbf{f}^\top \mathbf{U}_k \quad (24a)$$

$$\text{subject to} \quad -\mathbf{1}_{N_c} \bar{\delta}_r \leq \mathbf{U}_k \leq \mathbf{1}_{N_c} \bar{\delta}_r, \quad (24b)$$

where $\mathbf{H} := w_1 \gamma_r^{(r)\top} \gamma_r^{(r)} + w_2 \gamma_r^{(\beta)\top} \gamma_r^{(\beta)} + w_3 \mathbf{I}_{N_c} + w_4 \mathbf{D}^\top \mathbf{D}$, and $\mathbf{f} := w_1 \gamma_r^{(r)\top} (\phi^{(2,2)} r_k + \gamma_f^{(r)} \delta_{f,k} - \mathbf{r}_{d,k}) + w_2 \gamma_r^{(\beta)\top} (\phi^{(1,2)} r_k + \gamma_f^{(\beta)} \delta_{f,k}) - w_4 \mathbf{D}^\top \mathbf{u}_{\text{prev}}$. Any existing QP solver can be used to solve the problem in (24). Finally, the control input $\delta_{r,k}$ is obtained by applying the first element of the optimal solution $\mathbf{U}_k^*$ to the rear steering actuator.

V. VALIDATIONS

A. Validation Setup

The effectiveness of the proposed ARS control design was validated through numerical simulations in IPG CarMaker 15 [16]. The sampling time was set to 1 ms for the simulations, and 10 ms for the controllers, i.e., $T_s = 0.01$. The vehicle parameters used in the simulations, summarized in Table I, were the same as those used in the analyses in Section III. The longitudinal velocity of the vehicle was set to $v_x = 80 \text{ km h}^{-1}$. The driver's front steering input δ_f was generated using a Heaviside step function with an amplitude of 6.75 deg, which corresponds to a steering-wheel angle of 90 deg, while the rate limiter was set to 300 deg/s, as illustrated in Fig. 3a. The road friction coefficient was set to $\mu = 0.9$ for all simulations, which corresponds to dry asphalt road conditions.

To evaluate the sensitivity of conventional sideslip-angle-dependent controllers to imperfect sideslip-angle feedback, a corrupted feedback signal was introduced to represent both sensing noise and estimation errors. The feedback signal was modeled as

$$\beta_{\text{fb},k} = \beta_k + b_k + \mathcal{N}(0, (0.01 \text{ deg})^2), \quad (25)$$

TABLE II: Controllers and parameters used for comparative study.

Description and parameters	
(C ₁) [—]	Proposed MPC-based ARS: $N_p = 10, N_c = 2, \bar{\delta}_r = 3.5 \text{ deg},$ $w_1 = 185, w_2 = 10, w_3 = 50, w_4 = 1$
(C ₂) [—]	Conventional MPC-based ARS: same parameters as (C ₁)
(C ₃) [—]	STSMC-based ARS [7]: $w = -10, k_1 = 1.28, k_2 = 0.96$
(C ₄) [—]	Front-wheel steering only, i.e., $\delta_r = 0$

where $\beta_{\text{fb},k}$ denotes the sideslip-angle signal available to the controller and b_k denotes a random-walk error. The additive zero-mean Gaussian term $\mathcal{N}(0, (0.01 \text{ deg})^2)$ represents high-frequency sensing noise. The random-walk component represents slowly varying sensor bias or estimation drift and was generated by cumulatively summing Gaussian white-noise increments $\mathcal{N}(0, \sigma^2)$ at each time step, where σ denotes the standard deviation of each increment. To evaluate the sensitivity of the closed-loop performance to sideslip-angle feedback errors, simulations were conducted for $\sigma \in \{0, 0.0033, 0.0067, 0.01\} \text{ deg}$. Representative realizations of the sideslip-angle feedback error for $\sigma = 0 \text{ deg}$ and $\sigma = 0.01 \text{ deg}$ are shown in Fig. 3b.

For the comparative study, besides the proposed method (C₁), three baseline controllers were implemented, as summarized in Table II. The conventional MPC-based ARS controller is denoted by (C₂) and its MPC problem is formulated analogously to (18), but with the objective function designed to track the reference yaw rate and regulate the sideslip angle using the corrupted sideslip-angle feedback $\beta_{\text{fb},k}$, i.e., $\mathbf{x}_k = (\beta_{\text{fb},k} \ r_k)^\top$. Specifically, the objective functions J_1 and J_2 in (18a) (i.e., $w_1 J_1 + w_2 J_2$) are replaced with the following objective function:

$$J(\mathbf{U}_k) := \|\mathbf{X}_k - \mathbf{X}_{d,k}\|_{\mathbf{Q}}^2, \quad (26)$$

where $\mathbf{X}_{d,k} := \mathbf{1}_{N_p} \otimes (0 \ r_d(\delta_{f,k}))^\top$ is the reference state sequence along the prediction horizon, and $\mathbf{Q} := \text{diag}(\mathbf{1}_{N_p} \otimes (w_2 \ w_1)^\top)$ assigns w_1 to yaw-rate tracking and w_2 to sideslip-angle regulation, consistent with their roles in (C₁), where w_2 instead weights the suppression of the sideslip-dynamics residual.

Next, the STSMC-based ARS controller (C₃), which is a representative nonlinear control method for lateral vehicle control, was implemented based on the design in [7]. The sliding surface is defined as

$$s = w\beta + (r - r_d(\delta_f)), \quad (27)$$

where $w \in \mathbb{R}_{<0}$. Taking the time derivative of s , we have $\frac{d}{dt}s = \varphi_s + \gamma_s \delta_r$, where $\varphi_s := (w \ 1)(\mathbf{A}\mathbf{x} + \mathbf{b}_f \delta_f) - \frac{d}{dt}r_d$, $\gamma_s := (w \ 1)\mathbf{b}_r$. The STSMC control law, designed to drive the sliding surface s to zero, is given by

$$\delta_r = \frac{1}{\gamma_s} (-\varphi_s - k_1 |s|^{0.5} \text{sgn}(s) + z), \quad \frac{d}{dt}z = -k_2 \text{sgn}(s), \quad (28)$$

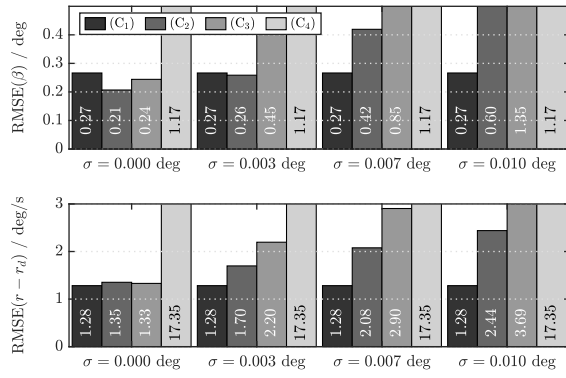


Fig. 4: Quantitative Comparison of Simulation Results.

where $k_1, k_2 \in \mathbb{R}_{>0}$.

Finally, (C₄) is a front-wheel-steering-only configuration that does not employ active rear steering, i.e., $\delta_{r,k} = 0$. All exact values of the parameters in (C₁)–(C₃) are summarized in Table II.

B. Validation Results

The root-mean-square error (RMSE) values of the sideslip angle and yaw-rate tracking error for controllers (C₁)–(C₄) under various feedback-error levels, $\sigma \in \{0, 0.0033, 0.0067, 0.01\}$ deg, are summarized in Fig. 4. As shown in the figure, both RMSE values of (C₁) remain unchanged regardless of σ , confirming that the proposed MPC design is independent of sideslip-angle feedback errors. In contrast, the RMSE values of both (C₂) and (C₃) increase with σ , indicating that their closed-loop performance is sensitive to sideslip-angle feedback errors. This increase is particularly pronounced for (C₃), whose RMSE values eventually exceed those of (C₄) as σ increases.

To further investigate these feedback-error-dependent variations, the time-domain responses for representative cases of $\sigma = 0$ deg and $\sigma = 0.01$ deg are illustrated in Fig. 5. First, as shown in Fig. 5a, (C₄) exhibits a large negative sideslip angle because no rear-wheel steering control is applied, indicating an oversteer tendency. In addition, its response remains unchanged regardless of the sideslip-angle feedback-error level.

On the other hand, for the case of $\sigma = 0$ deg, all controllers except (C₄) successfully suppressed the sideslip angle with ARS control, while tracking the reference trajectory of yaw rate. However, for the case of $\sigma = 0.01$ deg, (C₂) and (C₃) failed to suppress the sideslip angle, resulting in undesired tracking and stability performance of the vehicle, as shown in Fig. 5a. Accordingly, the yaw rate tracking performance of (C₂) and (C₃) was degraded, as shown in Fig. 5b. Conversely, (C₁) maintained consistent yaw-rate tracking and sideslip-angle suppression regardless of the sideslip-angle feedback-error level. Moreover, despite not using the sideslip-angle feedback, (C₁) achieved satisfactory control performance comparable to that of the conventional MPC-based ARS control (C₂) for the case of $\sigma = 0$ deg.

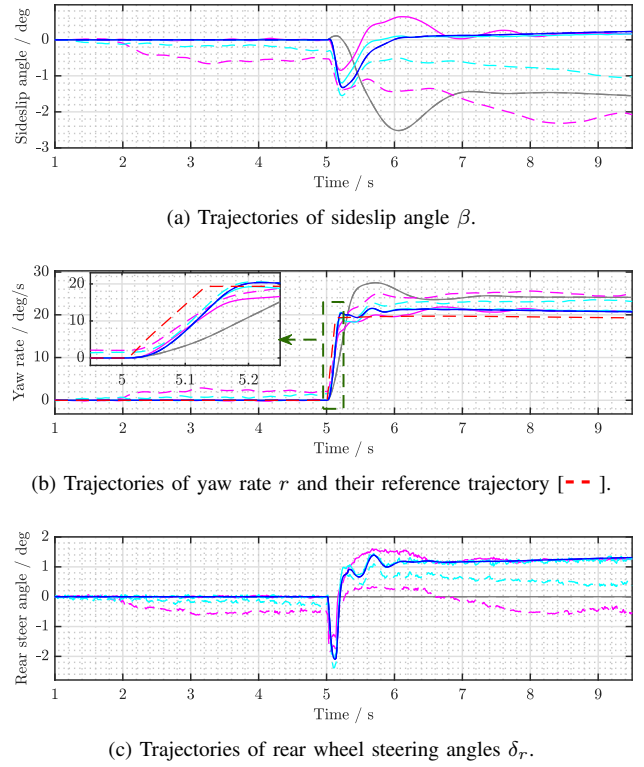


Fig. 5: Validation results of the proposed ARS control design for (C₁) [—], (C₂) [—], (C₃) [—], and (C₄) [—], with solid [—] and dashed [---] lines corresponding to $\sigma = 0$ deg and $\sigma = 0.01$ deg, respectively.

VI. CONCLUSION

In this paper, we proposed an MPC-based ARS controller that does not require sideslip-angle feedback. The proposed design eliminates the dependence on the sideslip angle through two complementary mechanisms: a sideslip-angle-independent yaw-rate predictor based on the limited finite-horizon influence of the initial sideslip angle, and a residual-suppression mechanism that promotes sideslip attenuation without directly using the sideslip angle. These mechanisms enable the MPC optimization to be formulated without either measured or estimated sideslip-angle feedback. Numerical simulations demonstrated that the proposed controller maintains satisfactory yaw-rate tracking and sideslip attenuation while remaining independent of sideslip-angle feedback errors. Future work will include real-time experimental validation and systematic tuning of the MPC weights over varying operating conditions.

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